

Chapter: Introduction

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Exercise 1.13

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

The maximum-likelihood variance estimator in (1.56) is

$$\sigma_{ML}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu_{ML})^2.$$

In this exercise, replace the estimate μ_{ML} by the true mean μ . Define

$$\tilde{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2.$$

We calculate its expectation:

$$\begin{aligned} \mathbb{E}[\tilde{\sigma}^2] &= \mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2 \right] \quad (\text{definition of } \tilde{\sigma}^2) \\ &= \frac{1}{N} \sum_{n=1}^N \mathbb{E} [(x_n - \mu)^2] \quad (\text{linearity of expectation}) \\ &= \frac{1}{N} \sum_{n=1}^N \text{Var}(x_n) \quad (\text{definition of variance around the true mean}) \\ &= \frac{1}{N} \sum_{n=1}^N \sigma^2 \quad (\text{each } x_n \text{ has variance } \sigma^2) \\ &= \sigma^2 \quad (\text{there are } N \text{ identical terms}). \end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}[\tilde{\sigma}^2] = \sigma^2}.$$

Thus, replacing μ_{ML} by the true mean μ removes the downward bias of the maximum-likelihood variance estimator.